1. The explaining process

Don't just tell people what the data say. Help them see what to notice and why it matters.

The big picture: three questions every explanation answers

1. **Describe:** What does the data show?
2. **Interpret:** What might it mean?
3. **So what?** Why does it matter to this audience? Say it in the plainest language you can.

Explaining a chart or table: four moves, then "So what?"

1. **Introduce:** "This graph shows..."
2. **Orient:** "The horizontal axis shows..." / "Each bar is..."
3. **Highlight:** "Look at... The main pattern is..."
4. **Interpret:** "This suggests that..."
5. **So what?** "In plain terms, this means..."

Choosing what to say

1. **Message:** What should the audience notice?
2. **Evidence:** The numbers that support the message.
3. **Detail:** More precise values, only if this audience needs them. Leave the rest on the screen.

Explaining a technical term

1. **Define:** "By *retention*, I mean how much people remember later."
2. **Rephrase:** "In other words..."
3. **Exemplify:** "For example, imagine a student who..."

The listener's three questions

- "What does that mean?"
- "Can you give me an example?"
- "So what?"

Adapted from Ken Bain, *What the Best College Teachers Do* (2004).

2. Example language

FUNCTION	EXAMPLE PHRASES
Averages	"The average score was 72." · "On average, participants scored..."
Proportions	"Approximately 60% of respondents..." · "Just over half of..." · "Just under one third of..." · "About three in ten..."
The denominator	"13 out of 40 students, about a third..." · "based on 1,054 respondents"
Comparisons	"Compared with Group B, Group A scored..." · "about five points higher than..." · "The difference was relatively small."
Change	"rose by 35" (the amount) · "rose to 840" (the new level) · "fell sharply to..." · "recovered to..."
Percentage points	"From 66% to 92% is a rise of 26 percentage points." · "That's an increase of about 39%." (Say which one you mean.)
Trends	increase · decrease · remain stable · fluctuate · peak · level off
Saying numbers aloud	54.2 = "fifty-four point two" · 115,146 = "about a hundred and fifteen thousand" · fif**TEEN** vs. **FIF**ty

FUNCTION	EXAMPLE PHRASES
Where to look	"Look at the line at the top..." · "On the right-hand side, you can see..." · "The orange bars show..." · "Notice the sharp drop in 2020."
Axes and honesty	"Note that the axis starts at 490, so the changes look bigger than they are."
Boxplots	"The line in the box is the median, the middle score." · "The box shows the middle half of the students." · "The dot is an unusual score." · "Scores in Group A were more spread out."
Careful claims	"The results suggest that..." · "This may be related to..." · "is associated with..." · "We can't say that ___ caused ___."
Limitations	"One limitation is that..." · "This should be interpreted with caution because..." · "We can't tell the reason from this chart alone."
Practical terms	"In practical terms, that's about three more words out of twenty." · "For a teacher, this means..."
So what?	"In plain terms, this means..." · "The key takeaway is..."

3. Useful frames

SITUATION	FRAME
Introducing yourself	"I'm ___ from the ___ department / office. I research / I work with data on ___."
Giving perspective	"___ out of ___, that's about ___%." · "Out of how many?"
Choosing a chart	"I'd use a ___ because it shows ___."
Checking you understood	"So the main point is..." · "Did you mean...?"
Explaining for a general audience	"For specialists, we'd say... In everyday terms, it means..."
A 90-second presentation	1. "We wanted to investigate..." 2. "This figure shows... Look at..." 3. "The most important finding is..." 4. "One possible explanation is..." 5. "So what? The key takeaway is..."
Giving feedback	"Your main point was clear." · "Could you tell me where to look?" · "What was the 'so what'?"